
Foreign Direct Investment and Economic Growth in Sub-Saharan Africa: Evidence from 2000-2024

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Abstract:

This study examines the relationship between foreign direct investment (FDI) and economic growth across Sub-Saharan African economies from 2000 to 2024. Using World Bank World Development Indicators, I analyze GDP per capita growth alongside FDI, gross capital formation, trade openness, and population growth. The analysis progresses from ordinary least squares to multivariate OLS and country fixed-effects models with clustered standard errors. The simple OLS estimate is positive but statistically insignificant. After adding controls and country fixed effects, the estimated FDI coefficient moves close to zero and remains statistically insignificant. A robustness specification excluding extreme FDI observations produces the same conclusion. Domestic investment and trade openness show stronger positive associations with growth in later specifications.

1 Introduction

Foreign direct investment is often viewed as an important source of capital for developing economies. It can bring financing, technology, managerial knowledge, employment, and access to international markets. For Sub-Saharan African economies, attracting FDI has therefore become an important component of many development strategies.

However, receiving foreign capital does not automatically imply faster economic growth. The benefits of FDI may depend on domestic

investment, trade integration, institutions, human capital, infrastructure, and the sectors receiving investment. This paper asks a focused empirical question: Is greater FDI associated with faster GDP per capita growth in Sub-Saharan Africa?

The analysis begins with the unconditional relationship between FDI and growth, then introduces economic controls and country fixed effects to test whether the relationship persists after accounting for observable factors and persistent cross-country differences.

2. Data

The analysis uses World Development Indicators from the World Bank for 48 Sub-Saharan African countries. The dependent variable is annual GDP per capita growth. The main explanatory variable is FDI net inflows as a percentage of GDP. Controls include gross capital formation as a percentage of GDP, trade as a percentage of GDP, and annual population growth.

Data availability differs across variables. GDP per capita growth has 1,169 observations, FDI has 1,162, gross capital formation has 1,013, trade has 1,019, and population growth has 1,199. Average FDI is approximately 4.26 percent of GDP and the median is approximately 2.51 percent. The distribution is highly skewed, ranging from about -17.3 percent to 103.3 percent of GDP.

3. Empirical Strategy

I estimate four specifications. Model 1 regresses GDP per capita growth on FDI using OLS with robust standard errors. Model 2 adds gross capital formation, trade openness, and population growth. Model 3 introduces country fixed effects and clusters standard errors by country, absorbing time-invariant country characteristics and focusing on within-country variation. Model 4 repeats the fixed-effects specification after excluding observations below the 1st percentile and above the 99th percentile of FDI.

The specifications identify statistical associations rather than causal effects. Reverse causality and omitted time-varying factors remain possible.

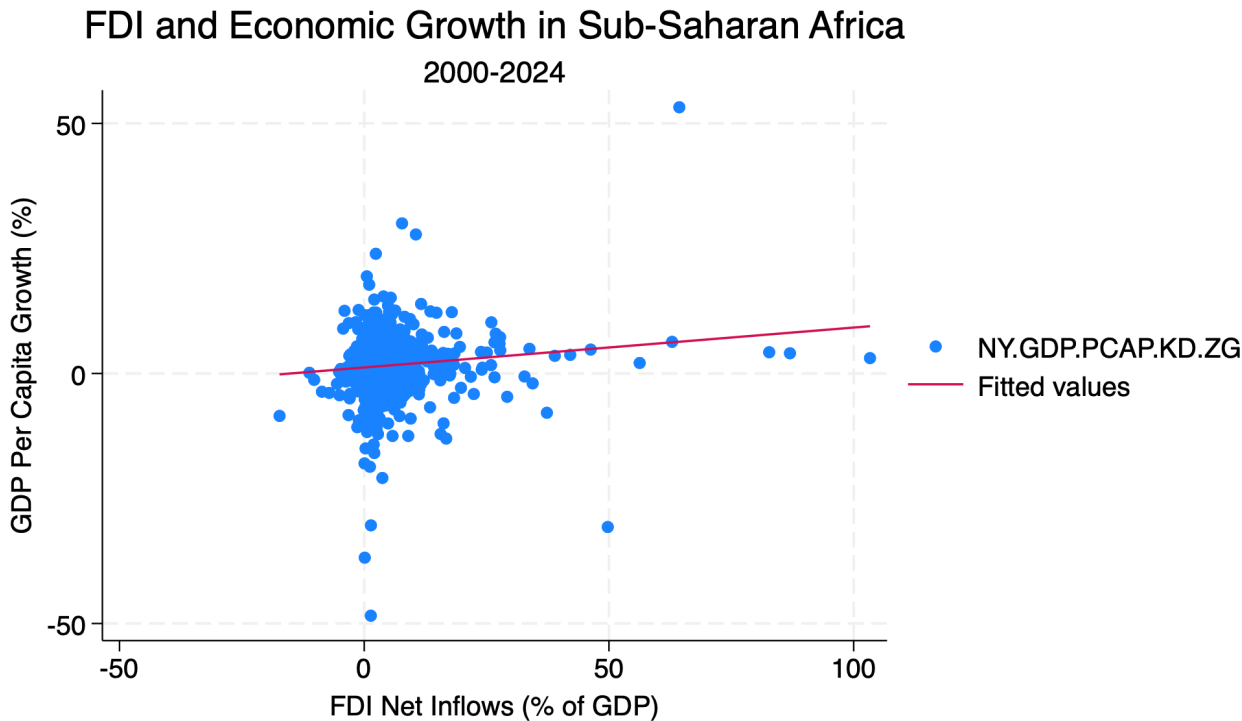


Figure 1. FDI and GDP per capita growth in Sub-Saharan Africa, 2000-2024.

4. Results

The simple OLS model produces an FDI coefficient of 0.0798. This implies that a one percentage-point increase in FDI as a share of GDP is associated with about a 0.08 percentage-point increase in GDP per capita growth in the unconditional specification. The estimate is not statistically significant.

After adding controls in Model 2, the FDI coefficient falls to 0.0041 and remains statistically insignificant. Gross capital formation has a coefficient of 0.116 and is statistically significant at the 1 percent level. In Model 3, the country fixed-effects estimate for FDI becomes -0.0131 and remains statistically insignificant. Trade becomes positive and statistically significant, with a coefficient of 0.0466.

5. Robustness

The 1st percentile of FDI is approximately -4.02 percent of GDP and the 99th percentile is approximately 34.42 percent, while the maximum exceeds 100 percent. Model 4 excludes observations outside this range. The FDI coefficient becomes 0.0126 and remains statistically insignificant.

Domestic investment is positive and statistically significant at the 5 percent level in the robustness specification, with a coefficient of 0.0869. Trade also remains positive and statistically significant, with a coefficient of 0.0450. The central FDI finding therefore does not appear to be driven by extreme observations.

Table 1. Regression estimates

Variable	OLS	OLS + Controls	Country FE	FE Robustness
FDI	0.0798 (0.0516)	0.00406 (0.0336)	-0.0131 (0.0384)	0.0126 (0.0631)
Investment		0.116*** (0.0287)	0.0831* (0.0435)	0.0869** (0.0423)
Trade		-0.00644 (0.00696)	0.0466** (0.0201)	0.0450** (0.0203)
Population growth		-0.448 (0.308)	-0.514 (0.489)	-0.826 (0.671)
Observations	1,162	995	995	978
R-squared	0.0140	0.0453	0.0450	0.0495

*Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

6. Limitations

The analysis is observational and should not be interpreted causally. Faster-growing economies may attract more foreign investment, creating reverse causality. Important time-varying factors such as political stability, institutional quality, commodity prices, education, infrastructure, exchange rates, and fiscal policy may affect both investment and growth. Aggregate FDI also masks differences across sectors and investment types.

7. Conclusion

This study finds little evidence of a robust statistical relationship between FDI inflows and GDP per capita growth across Sub-Saharan African economies from 2000 to 2024. The simple OLS estimate is positive but statistically insignificant. The relationship moves close to zero after adding controls and remains insignificant after introducing country fixed effects.

Removing extreme FDI observations does not change the conclusion. The findings do not imply that FDI is unimportant for development. Rather, they suggest that the relationship between foreign investment and growth is more complex than the volume of foreign capital entering an economy. Domestic investment, trade integration, institutional capacity, sectoral composition, and the interaction between foreign and domestic

capital may shape whether FDI contributes to sustained development.

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